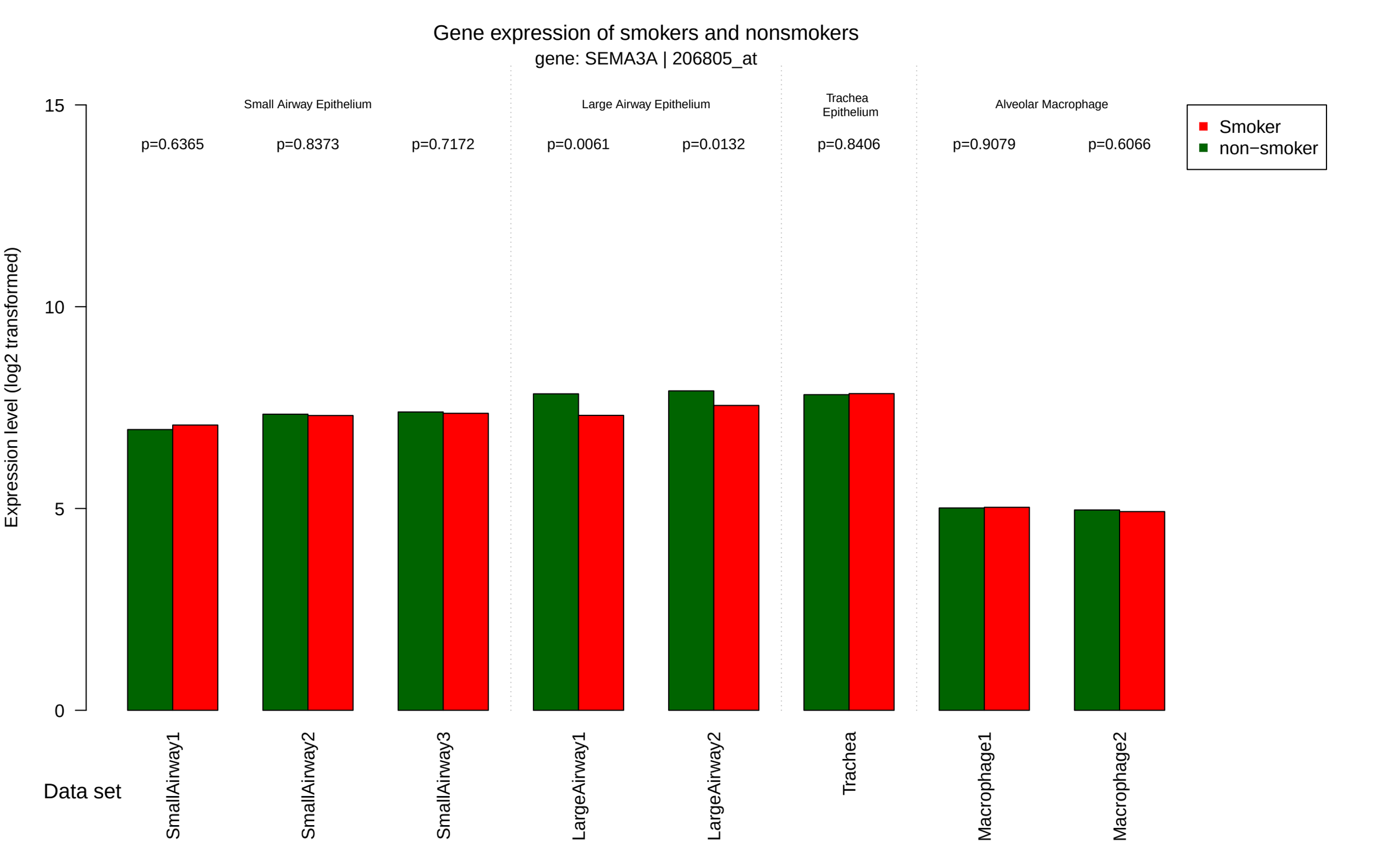
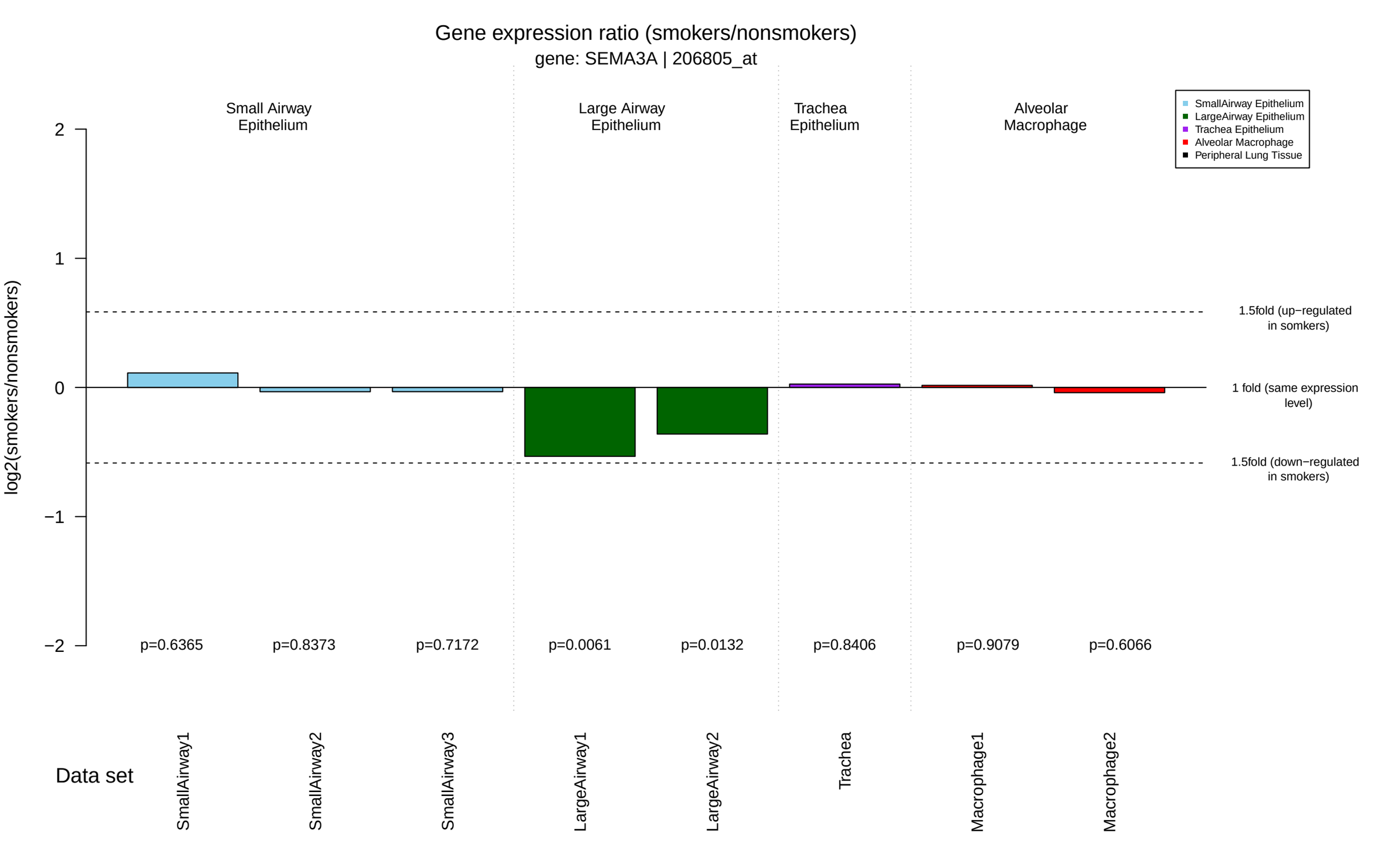
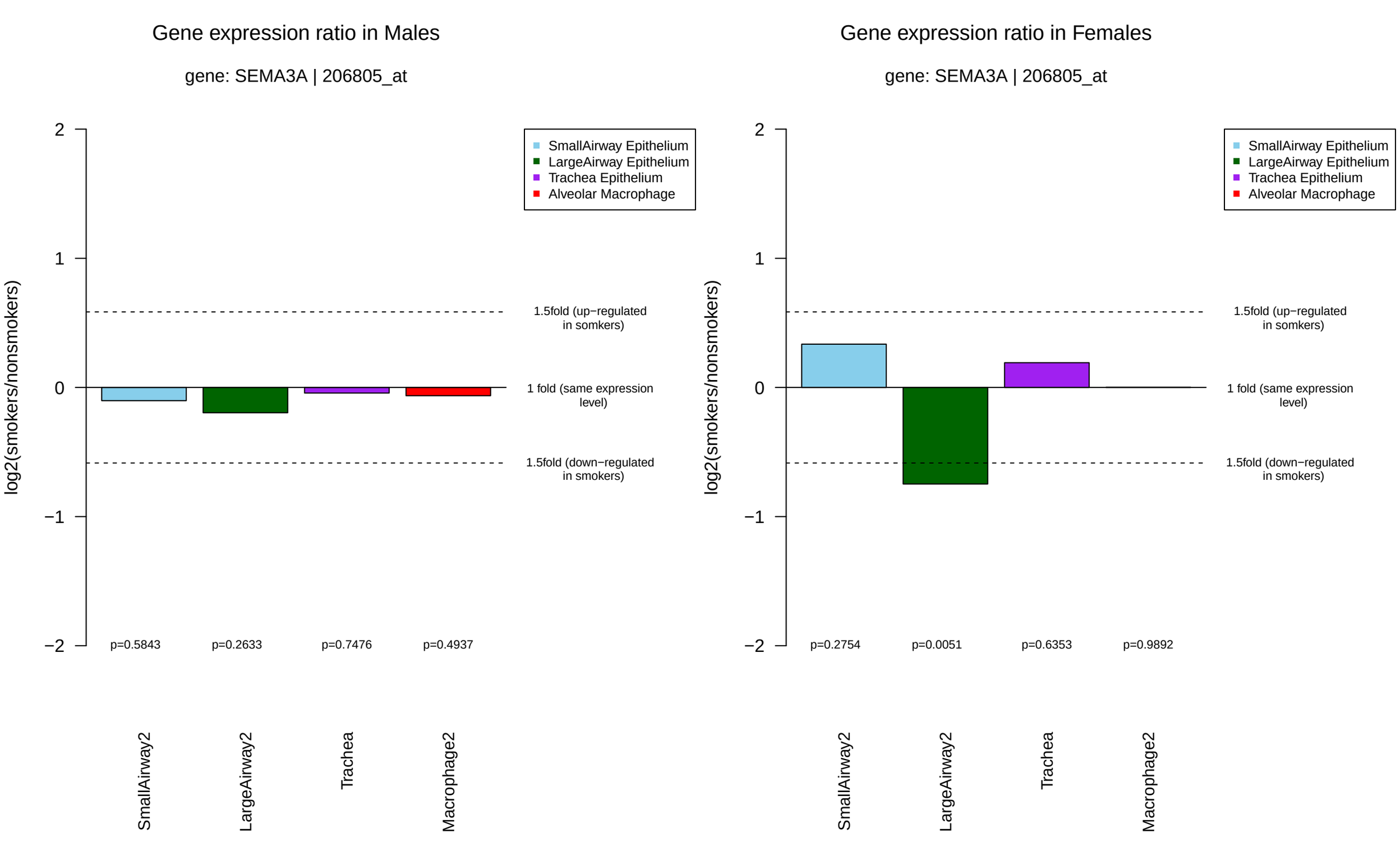
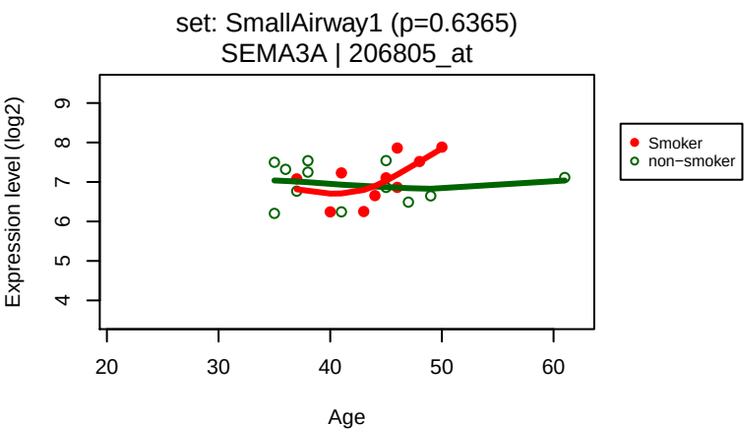
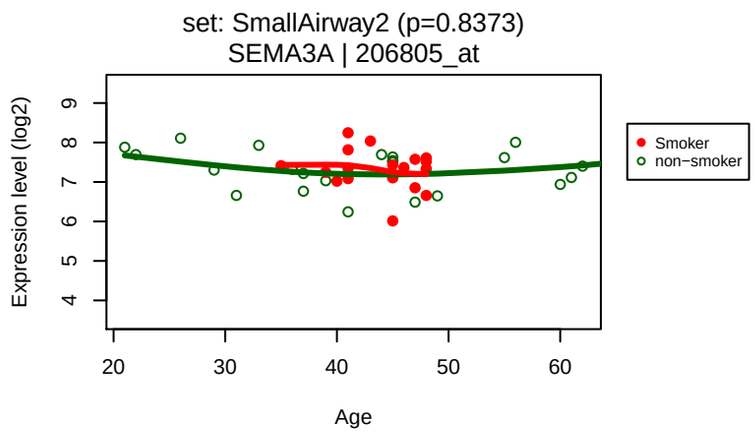
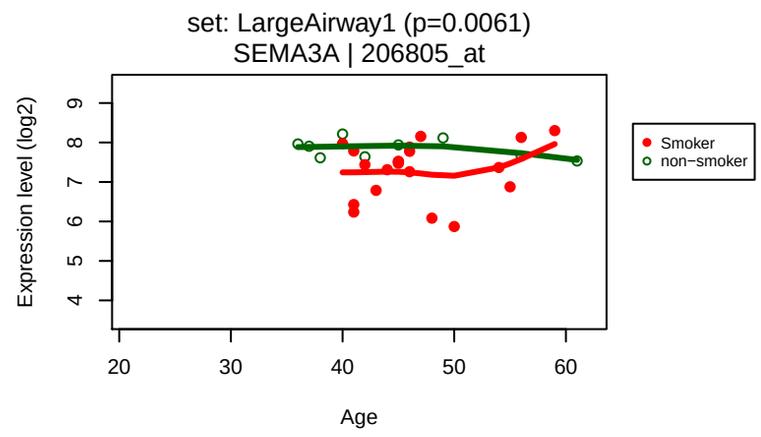

Gene expression of smokers and nonsmokers
gene: SEMA3A | 206805_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6365
p=0.8373
p=0.7172
p=0.0061
p=0.0132
p=0.8406
p=0.9079
p=0.6066
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

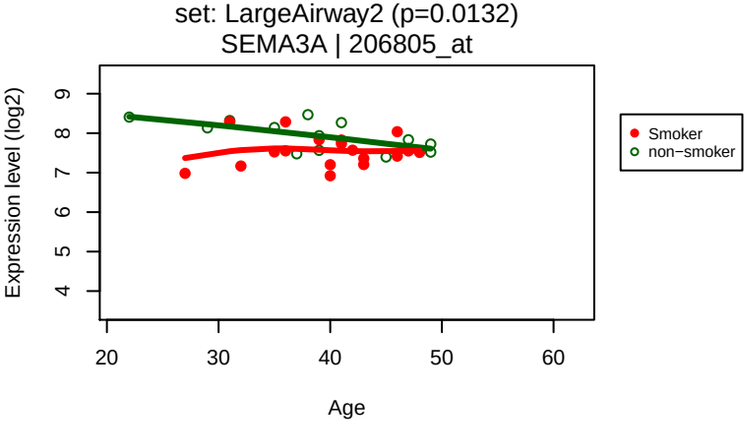

Gene expression ratio (smokers/nonsmokers)
gene: SEMA3A | 206805_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6365
p=0.8373
p=0.7172
p=0.0061
p=0.0132
p=0.8406
p=0.9079
p=0.6066
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

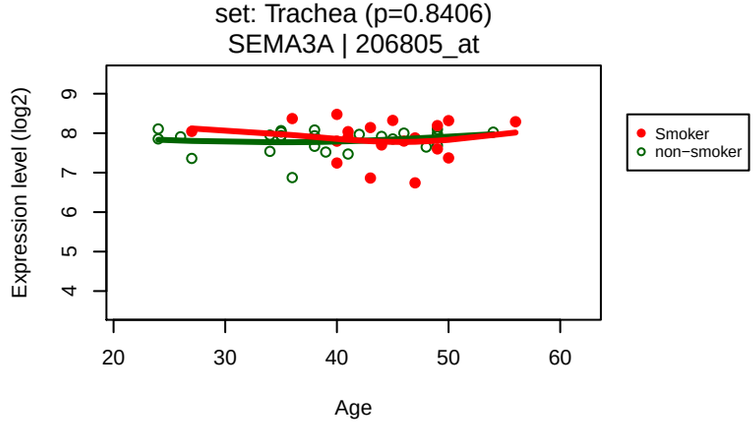

Gene expression ratio in Males
gene: SEMA3A | 206805_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5843
p=0.2633
p=0.7476
p=0.4937
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SEMA3A | 206805_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2754
p=0.0051
p=0.6353
p=0.9892
SmallAirway2
LargeAirway2
Trachea
Macrophage2

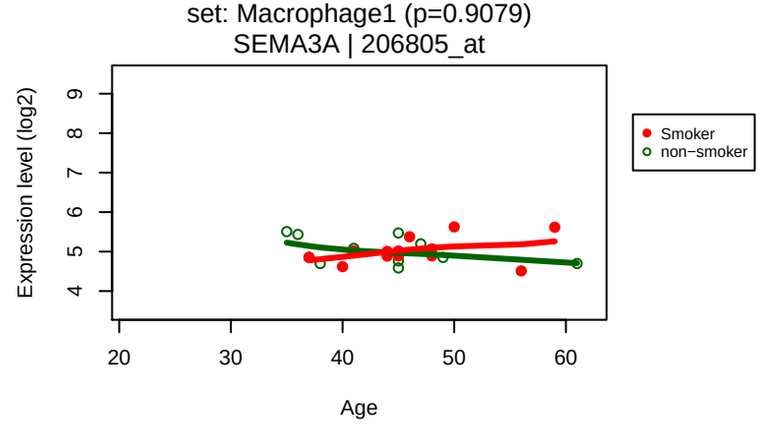

set: SmallAirway1 (p=0.6365)
SEMA3A | 206805_at
Expression level (log2)
Age
Smoker
non−smoker

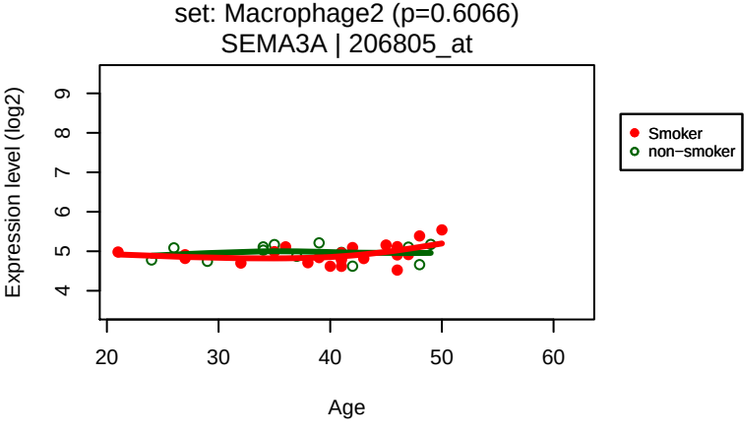

set: SmallAirway2 (p=0.8373)
SEMA3A | 206805_at
Expression level (log2)
Age
Smoker
non−smoker

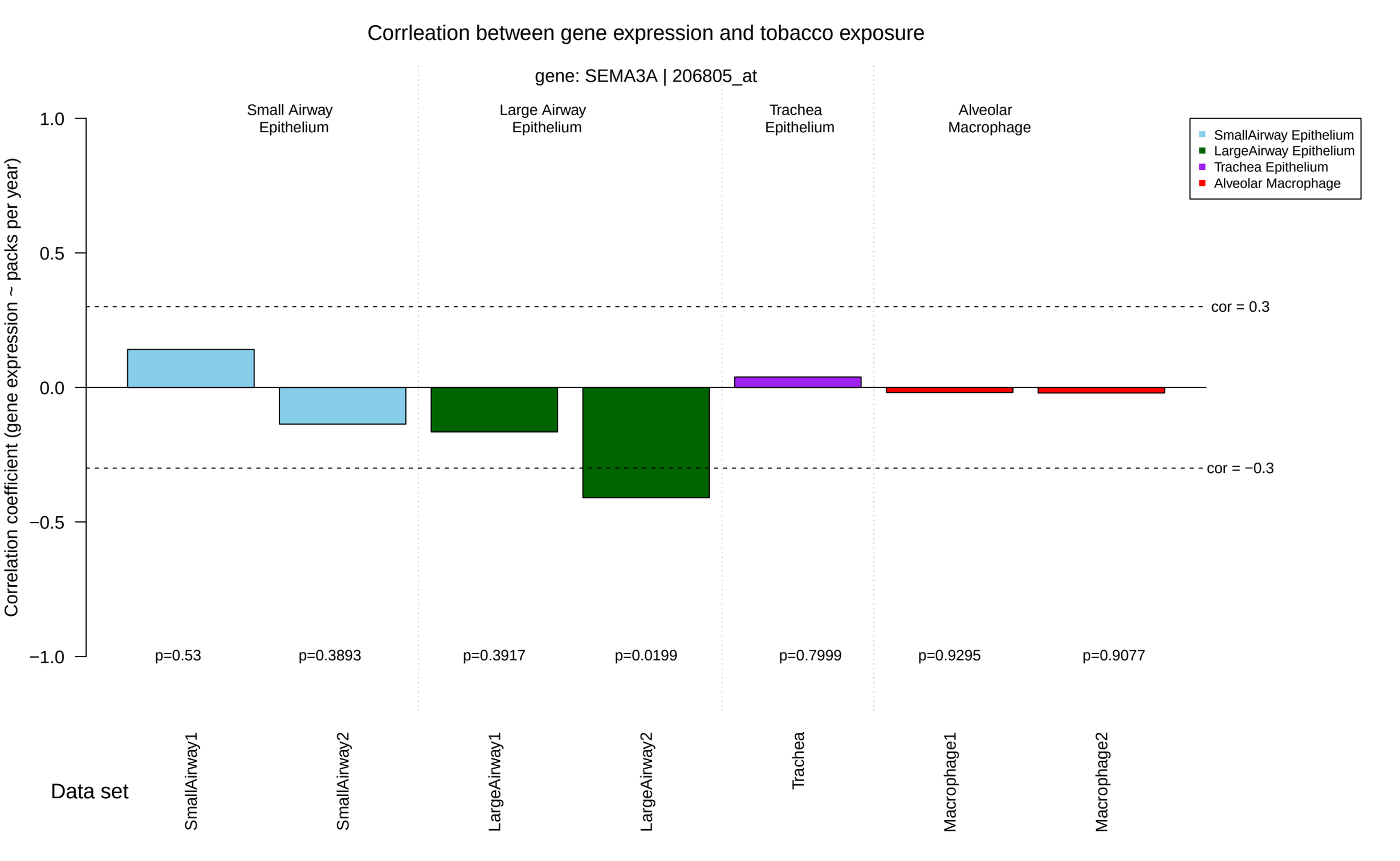

set: LargeAirway1 (p=0.0061)
SEMA3A | 206805_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0132)
SEMA3A | 206805_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8406)
SEMA3A | 206805_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9079)
SEMA3A | 206805_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6066)
SEMA3A | 206805_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SEMA3A | 206805_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.53
p=0.3893
p=0.3917
p=0.0199
p=0.7999
p=0.9295
p=0.9077
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2